

To avoid the problem of unnecessary multiple SMS being triggered just at geofencing border, often a buffer area is created around the geofencing border—say, 10 metres more in radius. When a user exits the geofencing border, an SMS is triggered. However, when the user enters the geofencing area and again exits, SMS is triggered only when the user travels the full 10 metres of buffer area within the geofencing boundary and then exits.

Also, one needs to be careful when using geofence for advertisement. Some people are likely to regard location tracking as something that bothers them and infringes on their privacy.

When integrated with CRM, geofence helps to collect useful data. You can add value to your app and your business as you are able to come up with something timely, relevant and useful. Your geofence messages are likely to improve your customers' experience and not flagged as spams.

Other applications

Geofences can also be used to keep an eye on things (know if a vehicle or piece of equipment is removed from your yard) or to keep things out (for example, to keep vehicles out of a restricted area, or to keep mining workers out of an area where explosives are being used). Today, it is not easy to manage on-the-ground workforce. However, it can be streamlined with the help of route optimisation techniques and map-based real-time tracking.

Geofencing acts as a great facilitator. You can demarcate delivery fences and, by providing your workforce with smartphones, receive notifications every time a delivery goes off-track. Moreover, if delivery packages are equipped with RFID tags, these too can be tracked throughout the delivery process.

In case your business depends on collaborating with other local businesses, a geofence app based interface



Fig. 4: Geofencing in marketing (Source: <https://blog.beaconstac.com>)



Fig. 5: A message announcing a coupon pops up on the mobile screen via geofencing (Source: www.hongkiat.com)



Fig. 6: A sample of geofencing based report for vehicle identification (Source: www.randmccally.com)

will make it possible for you to be aware of and communicate with local retailers, delivery warehouses as well as local operators. Thus geofencing facilitates expansion of a centralised business.

Another application is getting more out of your analytics program.

When used in conjunction with location-based information, analytic data generates geofenced metrics. These enable you to understand behaviours like the time spent at various locations, purchase preferences, area preference and time spent on different kinds of shopping. You can even better understand the effectiveness of your promotional messages and accordingly fine-tune them.

Geofencing can also be useful in event organisation. Geofence becomes a virtual perimeter around a physical venue—perhaps a conference hall area in a hotel. It then communicates with app users (delegates) while they are inside it. By using their phones' GPS locator and a transmitter such as iBeacon, conference organisers can help attendees who walk in through the door to smoothly check-in along with opt-in social media networking and directions to specific points of interest. The event-specific app will inform attendees whether a certain colleague is present at the same venue and even allow them to share contact details.

Participants at a tradeshow can stream demo videos to potential clients as they pass by their stall, or offer e-coupons to those who seem keenly interested. Such event-based apps also help to provide attendees with information on who is speaking, when and where. Seamless two-way communication is also possible—for example, audience can key in questions for a speaker and send them off to be archived and answered when time permits.

Geofencing is being used to restrict firearms only to permitted areas, thereby rendering them unusable in restricted areas. In some firms, it is used by the human resource (HR) department to keep an eye on employees working in special locations, more so while on field duty. The geofencing tool enables employees to log their attendance with the help of a GPS-